

MARIHOT JOSUA TAMBUNAN

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Balige, North Sumatera, Indonesia

Fresh Graduate in Applied Software Engineering Technology with a strong focus on Data Analytics, Data Engineering, and Machine Learning. Proficient in SQL, Python, Tableau, and Power BI, with practical experience in end-to-end data modeling and analytics.

JENJANG PENDIDIKAN

Institut Teknologi Del - Laguboti, North Sumatera, Indonesia
Bachelor of Applied Software Engineering Technology, 3.35/4.00

Aug 2022 - Sep 2026

WORK EXPERIENCE

PT Sembrani Siber Solusindo - Remote
Penetration Tester Intern

May 2026 - Sekarang

An ICT solutions and cybersecurity firm specializing in enterprise technology, consulting, and cyber incident management

- Conducted full-scope security assessments and vulnerability evaluations across client web applications and network infrastructure
- Identified critical attack vectors and unauthorized access risks using industry-standard penetration testing methodologies
- Authored actionable security assessment reports and mitigation strategies to strengthen client cyber defenses and data integrity

PT Wide Technologies Indonesia - Jakarta, Indonesia
Software Engineer Intern

Jul 2025 - Dec 2025

A company that provides financial technology (fintech) solutions and IT infrastructure

- Analyzed enterprise banking CMS architecture (Primecash) built on Java (Spring MVC, Hibernate ORM, JDBC) to evaluate system workflows and data integrity
- Audited black-box Testing using OWASP ZAP, Burp Suite, and Postman to detect BAC, SSRF, and Auth failures
- Authored technical security reports and built a Laravel 10 application demonstrating secure coding remediations

Coding Camp by DBS Foundation - Remote
Machine Learning Engineer Intern

Jan 2025 - Jun 2025

Gained hands-on experience in data science, data analytics, and machine learning through an intensive coding bootcamp

- Proficient in SQL and Python for data analysis using Pandas, NumPy, and Matplotlib
- Implemented Machine Learning and basic Deep Learning models from theory to practice
- Applied end-to-end workflows: data preprocessing, feature engineering, model evaluation, and deployment

PROJECTS

Comparative Analysis of Child Activity Classification with Limited Dataset - Toba, Indonesia

Jan 2026 - Jul 2026

Undergraduate thesis research

- Built an end-to-end data pipeline to clean, standardize, and augment 4,800+ images across 4 activity classes
- Optimized a CNN-SVM model using Grid Search and 5-Fold CV, achieving 87.41% accuracy and 0.8746 F1-Score

Development of Agricultural Potential Data Collection Application for North Tapanuli Region Web-based System - Toba, Indonesia

Jan 2025 - Jun 2025

3rd-Year Final Project (PA 3)

- Architected a normalized PostgreSQL database (15+ entities) to unify regional agricultural and price tracking data
- Built SQL data aggregation pipelines (~21k LOC) and automated analytical dashboards across 60+ validated test scenarios

MediVize: Web-Based Drug Information System with Image Classification - Indonesia

Apr 2025 - Jun 2025

DBS Foundation Capstone Project

- Executed data cleaning and NumPy-based image augmentation across 150 drug categories to optimize training data quality
- Trained a YOLOv11 detection model achieving 96.5% mAP@50 (87.3% Precision, 93.4% Recall) for automated drug identification

Model - Toba, Indonesia

Machine Learning Course Final Project

- Engineered a hybrid ML pipeline using K-Means, SMOTE, and StandardScaler to clean and rebalance student cognitive datasets
- Fine-tuned SVM and Random Forest models via GridSearchCV, achieving 97.67% and 97.42% accuracy for adaptive learning

PENGALAMAN ORGANISASI

MPM - Del Institut of Technology

Sep 2024 - Jan 2026

Head of Public Relations, Publication and Career Commission

Majelis Permusyawaratan Mahasiswa

- Led strategic communications and digital media to elevate organizational branding and engagement
- Coordinated career development programs, networking events, and internal documentation systems

KPU - Del Institut of Technology

Apr 2024 - Nov 2024

Member of Security and Infrastructure Division

Komisi Pemilihan Umum

- Managed security protocols and infrastructure systems for safe and transparent campus election processes
- Coordinated facility management and logistical support for election activities and venue preparation

HIMAPRO - Del Institut of Technology

Sep 2023 - Sep 2024

Member of Event Division

Himpunan Mahasiswa Teknologi Rekayasa Perangkat Lunak

- Served as Master of Ceremonies for departmental events to drive student engagement
- Co-designed event concepts, managed budgets, and coordinated cross-divisional teams for seamless execution

GDSC - Del Institut of Technology

Sep 2023 - Sekarang

Member

Google Developer Student Club

- Worked with peers on group projects, applying Google technologies to solve real-world challenges
- Engaged in hands-on sessions, such as coding workshops and project-building, to strengthen technical skills

CERTIFICATION & SKILLS

- ✓ Certification (2025): TOEFL ITP (Score : 513) | ETS
- ✓ Certification (2025): Machine Learning Terapan | Dicoding
- ✓ Certification (2025): Belajar Dasar Visualisasi Data | Dicoding
- ✓ Certification (2025): Belajar Analisis Data dengan Python | Dicoding
- ✓ Certification (2025): Belajar Fundamental Deep Learning | Dicoding
- ✓ Certification (2025): Belajar Machine Learning untuk Pemula | Dicoding
- ✓ Certification (2024): Data Science Project: Predictive Analytics | Eduwork.id (PT Talenta Sinergi Group)
- ✓ Certification (2024): Python Fundamental for Data Science | DQlab
- ✓ Certification (2024): Data Engineering Project: Taxi Travel Data Analysis | Dibimbing.id

Data Analytics & Visualization: Tableau, Power BI, Advanced Excel, Exploratory Data Analysis (EDA)

Databases & Big Data: PostgreSQL, MySQL, MS SQL Server, MongoDB, Hadoop, Cassandra

Machine Learning & Data Science: Python, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV

Programming & Developer Tools: Python, SQL, Java, Git, RESTful API, Streamlit

Soft Skills: Problem Solving, Communication Skills, Adaptability, Time Management, Teamwork, Attention to Detail, Analytical Thinking, Curiosity and Continuous Learning